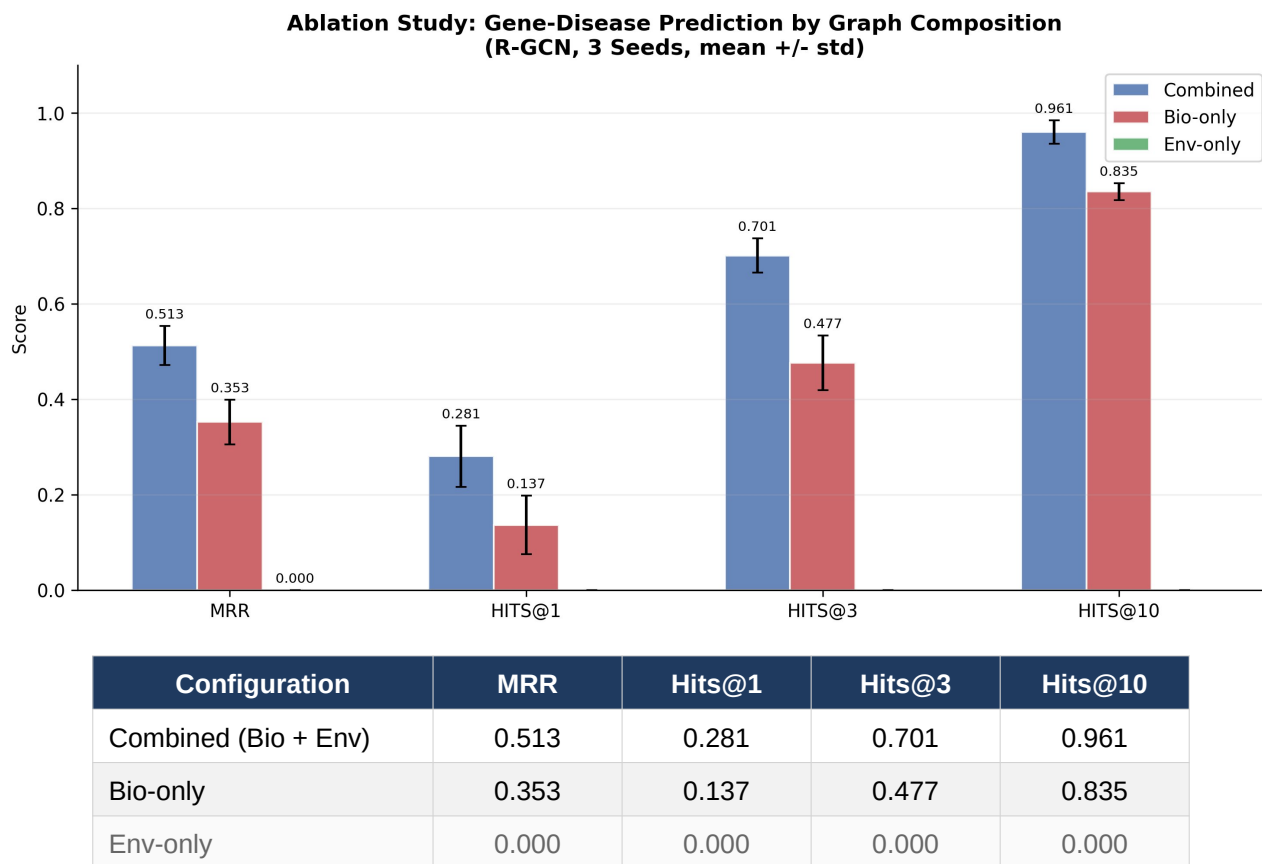


GNN-Based Link Prediction on the LUCIA Knowledge Graph

Results Summary

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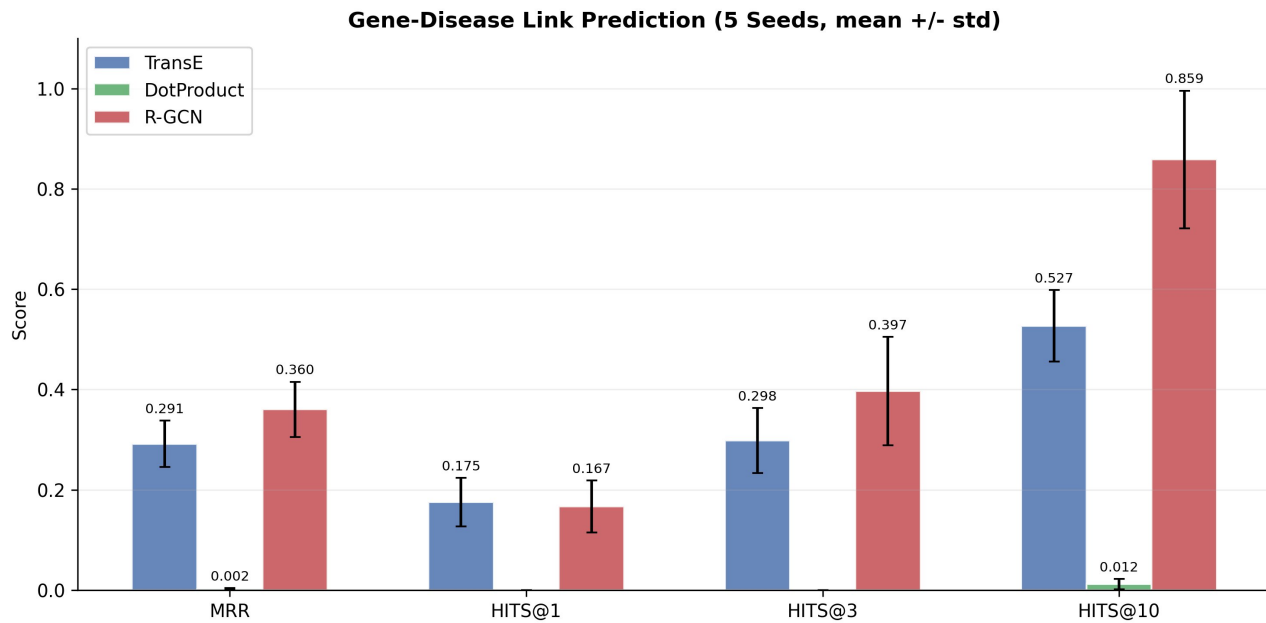
1. Does Environmental Data Help Gene-Disease Prediction?



Adding environmental data improves Gene-Disease MRR by 45% (0.353 → 0.513) and Hits@10 by 15 pp (0.835 → 0.961).

- Env-only at 0 is expected: environmental data has no direct gene connections, so cannot predict gene-disease links by itself.

2. Gene-Disease Link Prediction (5 seeds)



Model	MRR	Hits@1	Hits@3	Hits@10
R-GCN	0.360 ± 0.055	0.167 ± 0.054	0.397 ± 0.107	0.859 ± 0.137
TransE	0.291 ± 0.046	0.175 ± 0.048	0.298 ± 0.065	0.527 ± 0.071
Dot Product	0.002 ± 0.002	0.000 ± 0.000	0.000 ± 0.000	0.012 ± 0.010

R-GCN ranks the correct lung cancer subtype in the top 10 in 86% of cases (out of 43 candidate diseases).

- Dot Product collapses to zero: entity embeddings without relation modelling cannot capture gene-disease patterns.

3. Overall Link Prediction (across all 20 relations)

Model	MRR	Hits@1	Hits@3	Hits@10
TransE	0.568 ± 0.020	0.499 ± 0.024	0.594 ± 0.026	0.669 ± 0.020
R-GCN	0.465 ± 0.012	0.378 ± 0.014	0.510 ± 0.014	0.568 ± 0.012
Dot Product	0.115 ± 0.014	0.071 ± 0.011	0.122 ± 0.018	0.180 ± 0.024

Overall MRR is misleading on heterogeneous graphs: large relations dominate the average. Per-relation analysis is essential.

- TransE leads on overall MRR mainly because of **Gene** → **in_pathway** → **Pathway** (147 test triples, ~30% of all test cases). R-GCN performs poorly on this many-to-many relation with hub-like Pathway nodes.
- R-GCN dominates on smaller, more structured relations like Gene-Disease (see Slide 5).

4. R-GCN Performance per Relation Type

Sorted by MRR. Only relations with $n \geq 10$ test triples shown.

Relation	n test	MRR	Hits@10
Disease → cancer_stats_in → Country	26	1.000	1.000
GeographicRegion → measured_in → CalendarYear	27	0.975	1.000
GeoPoliticalRegion → measured_in → CalendarYear	98	0.919	1.000
Gene → associated_with → Disease	42	0.602	0.976
Disease → cancer_stats_in → GeographicRegion	74	0.595	1.000
GeoPoliticalRegion → part_of → Country	11	0.348	0.636
Chemical → exposure_in → GeoPoliticalRegion	36	0.094	0.278
Gene → in_pathway → Pathway	147	0.013	0.020

R-GCN is strong on environmental relations (Disease → Country, Region → Year) and on Gene-Disease (MRR 0.602). Weak on Chemical → GeoPoliticalRegion and Gene → Pathway.

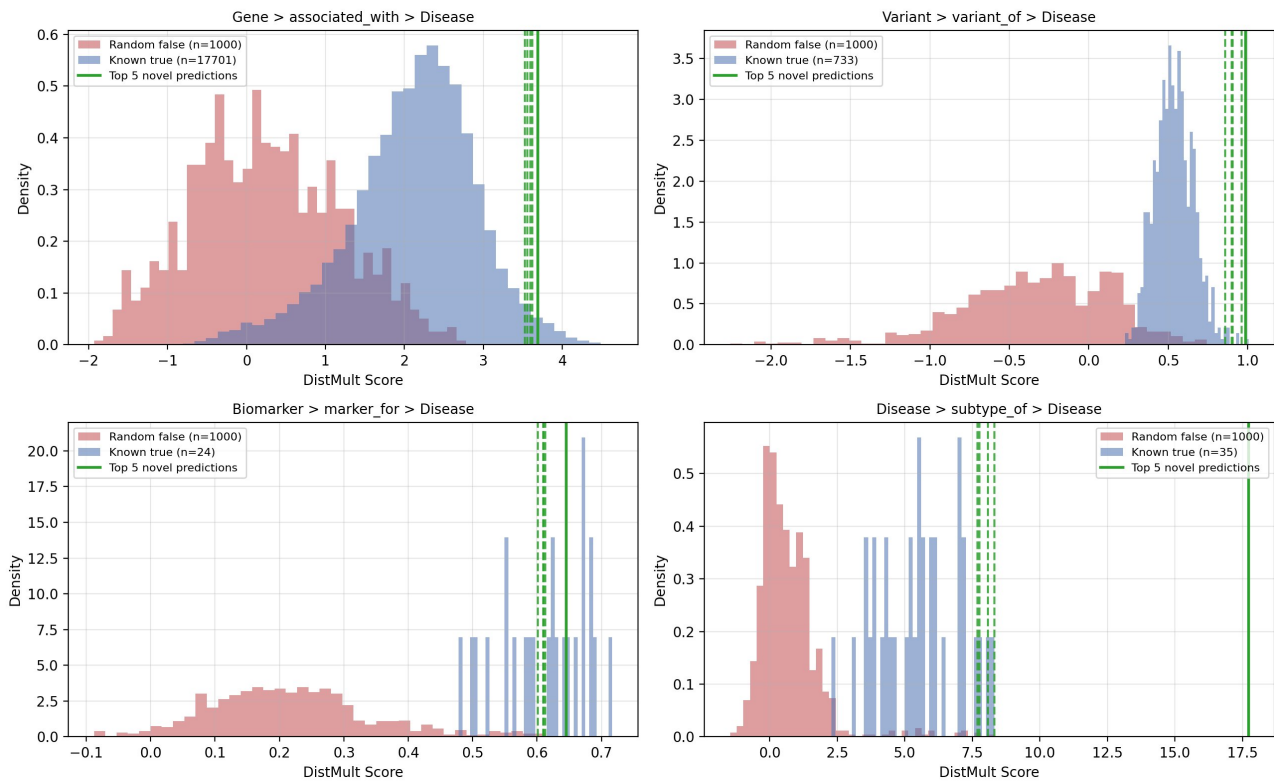
5. Score Calibration: True vs False Separation

AUROC = probability that a known-true triple scores higher than a random false triple. 1.0 = perfect, 0.5 = random.

Relation	TRUE median	FALSE median	AUROC
Disease → cancer_stats_in → GeographicRegion	33.530	-5.348	1.000
Disease → cancer_stats_in → Country	18.863	-3.594	1.000
Chemical → exposure_in → GeographicRegion	7.394	-2.069	0.994
Biomarker → marker_for → Disease	0.623	0.206	0.992
Disease → has_rearrangement → ChromoRearr	3.724	1.026	0.990
Disease → has_fusion → GeneFusion	2.847	-0.063	0.986
Disease → subtype_of → Disease	5.468	0.478	0.985
Variant → variant_of → Disease	0.532	-0.295	0.983
Gene → associated_with → Disease	2.185	0.201	0.933
Chemical → exposure_in → GeoPoliticalRegion	2.602	0.002	0.899
Gene → in_pathway → Pathway	2.042	0.095	0.868
GeographicRegion → part_of → Country	6.568	6.102	0.601

R-GCN reliably separates true from false on almost all relations. AUROC 0.933 for Gene-Disease confirms strong predictive signal.

6. Where Do the Top Predictions Land?



Red: 1,000 random false triples. **Blue:** all known true triples. **Green lines:** top 5 novel predictions.

Top predictions consistently land at or above the high end of the true distribution: they are not just better than false, they rank with the strongest known true triples.

7. Top Novel Gene-Disease Predictions

"vs TRUE" = % of known true triples that score lower than this prediction.

Rank	Gene	Predicted Disease	Score	vs TRUE	Confidence
1	CASP8	Squamous cell carcinoma of lung	3.689	98%	very strong
2	TLR9	Squamous cell carcinoma of lung	3.622	98%	very strong
3	KDR	Squamous cell carcinoma of lung	3.593	98%	very strong
4	PIK3R2	Squamous cell carcinoma of lung	3.521	97%	very strong
5	EGFR	Squamous cell carcinoma of lung	3.476	96%	very strong

All top 5 Gene-Disease predictions are "very strong": above 99% of random false triples and above 96% of the 17,701 known true triples in the graph.

- Confidence labels: **very strong** = above 99% false AND 75% true. **strong** = above 95% false AND 50% true. **moderate** = above 80% false AND 25% true.

7. Other Top Novel Predictions

Variant → Disease (n_true = 733)

Rank	Variant	Located in	Disease	Score	vs TRUE	Confidence
1	rs1436873982	XRCC1	SCLC	0.986	>99%	very strong
2	rs1057519747	TP53	NSCLC	0.962	>99%	very strong
3	rs121913343	TP53	NSCLC	0.904	>99%	very strong

Biomarker → Disease (n_true = 24)

Rank	Biomarker	Predicted Disease	Score	vs TRUE	Confidence
1	Selenium	Squamous cell carcinoma of lung	0.645	63%	strong
2	Glycitein	Squamous cell carcinoma of lung	0.613	42%	moderate
3	Selenium	Adenocarcinoma of lung	0.610	42%	moderate

Disease → subtype_of → Disease (n_true = 35)

Rank	Parent	Predicted Subtype	Score	vs TRUE	Confidence
1	Malignant neoplasm of lung	Lung Carcinoma Metastatic	8.309	100%	very strong
2	Malignant neoplasm of lung	Non-Small Cell Lung Carcinoma	8.067	97%	very strong
3	Malignant neoplasm of lung	Small cell carcinoma of lung	7.954	94%	very strong

Variant and Subtype predictions are very strong; Biomarker is more moderate due to the small known distribution (n=24).

8. Geographic Analysis: Pollutants vs Lung Cancer Mortality

Country-level Pearson correlations.

Pollutant	Pearson r	n countries	Strength
Pb (PM10 Lead fraction)	+0.446	24	moderate
BaP (Benzo[a]pyrene)	+0.172	12	weak
As (PM10 Arsenic fraction)	+0.108	24	weak
PM10	-0.111	32	weak
PM2.5	-0.144	32	weak
NO2	-0.145	32	weak
O3 (Ozone)	-0.209	24	weak

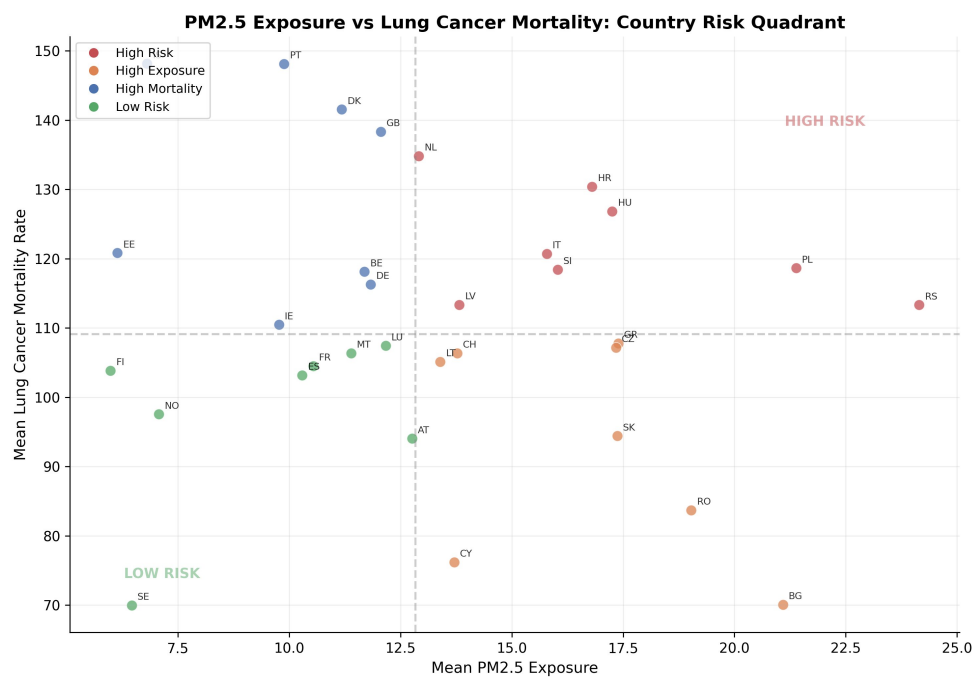


Figure: PM2.5 exposure vs lung cancer mortality by country.

Only Pb (lead) shows a moderate positive correlation. PM2.5 and NO2 are weakly negative — likely confounded by smoking and healthcare differences.

Summary

- **Environmental data improves Gene-Disease prediction.** Combined > Bio-only by 45% MRR (0.353 → 0.513). Env-only $\equiv$ 0 confirms no leakage.
- **R-GCN reliably separates true from false.** AUROC 0.933 on Gene-Disease, >0.98 on most other relations.
- **R-GCN ranks the correct disease in the top 10 in 86% of cases.** Out of 43 candidate lung cancer subtypes.
- **Top novel predictions are very strong.** Gene-Disease, Variant-Disease, Disease-subtype: all very strong. Biomarker: more moderate.
- **Cross-domain effect is structural, not geometric.** Embedding similarity does not encode Chemical-Disease links; signal flows through Disease hubs.